

3. Failure: Creep

1. **Creep** is a slow, continuous deformation with time at elevated temperature.

$$\varepsilon = f(\sigma, t, T)$$

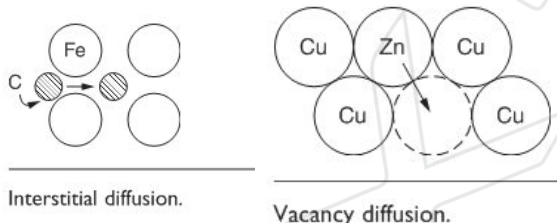
For creep, the actual temperature is not important. It is the homologous temperature T/T_m that matters. Creep occurs at $T > \sim 0.3 T_m$ in metals (Temperature in K, not °C)

Since T_m is high for most aerospace alloys, creep occurs at elevated temperature.

2. **Diffusion mechanisms**

Interstitial atoms can jump from interstitial site to interstitial site provided they have enough energy to overcome the barrier, q .

Substitutional atoms require both sufficient energy and for there to be a vacant lattice site to jump into. So, interstitial diffusion is much faster.



Grain boundaries can act as fast diffusion paths. If the grain size is very small, grain boundary diffusion can be important.

Dislocation cores can act as fast diffusion 'wires'. If there is a high dislocation density, diffusion along dislocation cores can be important.

3. **Two main creep mechanisms**

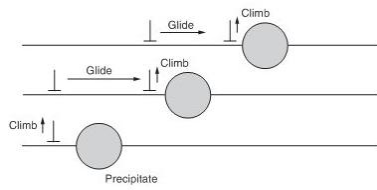
There are two main creep mechanisms: dislocation creep and diffusion creep.

a. Dislocation creep

Dislocation glide: dislocations moving conservatively along their slip planes. It occurs if the stress is large enough for the dislocations to overcome the lattice resistance and obstacles in the lattice.

Dislocation climb: movement of dislocations outside their glide plane by diffusion of vacancies to the bottom of the half plane. It occurs only at relatively high temperatures when diffusion becomes significant.

Dislocation creep: the combination of slip and jump allows dislocations to become 'unlocked' from obstacles such as: precipitates, solute atoms and other dislocations

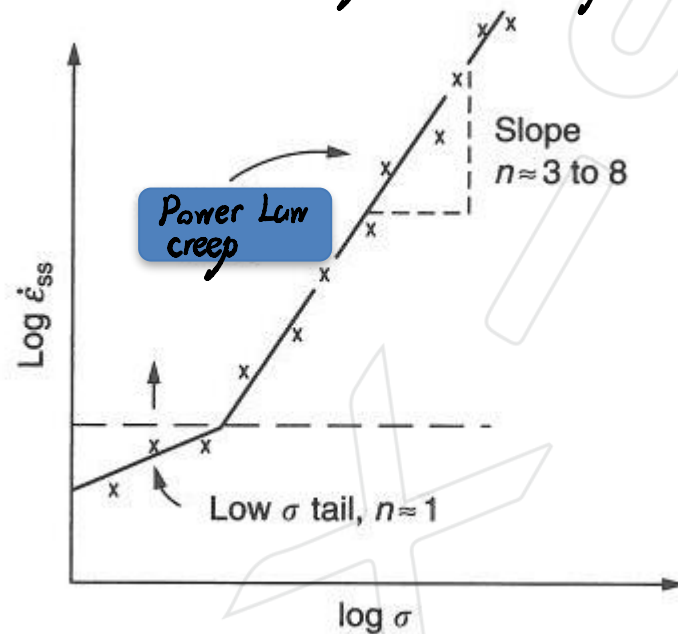


How the climb-glide sequence leads to creep.

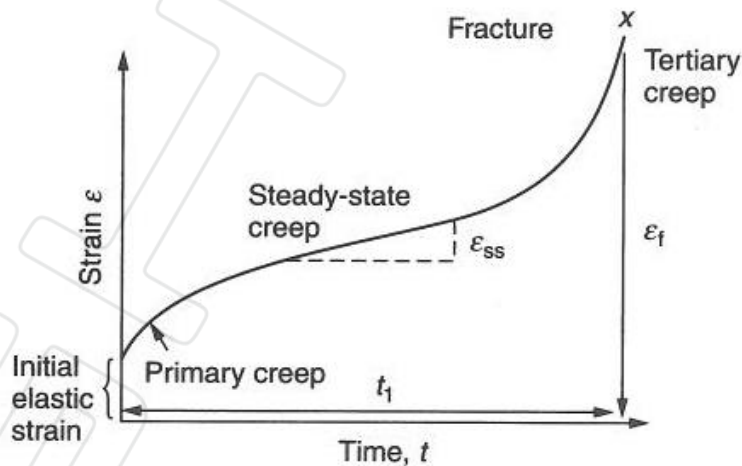
The factors influencing the rate of dislocation creep are: stress
and temperature.

$$\dot{\epsilon} = A\sigma^n e^{-\frac{Q}{kT}}$$

Dislocation creep results in power law creep.



Power law creep only occurs at steady state and at relatively high stress levels. As shown in figure below, the extension is measured as a function of stress. Primary creep and initial elastic strain occurs very fast. Then the material enters steady state creep and the strain increases steadily with the time. This part of curve is the most important for design.



b. Diffusion creep

At relatively low stress, power law creep falls quickly. Shape change occurs by diffusion of the atoms. Creep happens by atoms and vacancies rearranging parallel to the loading direction (dislocations are not involved).

At low temperatures the diffusion occurs mainly along easier paths, such as grain boundaries, where the number of vacant atom sites is high. This is known as Coble creep.

At high temperatures, atoms and vacancies diffuse through the bulk of the material. This is known as Nabarro - Herring creep.

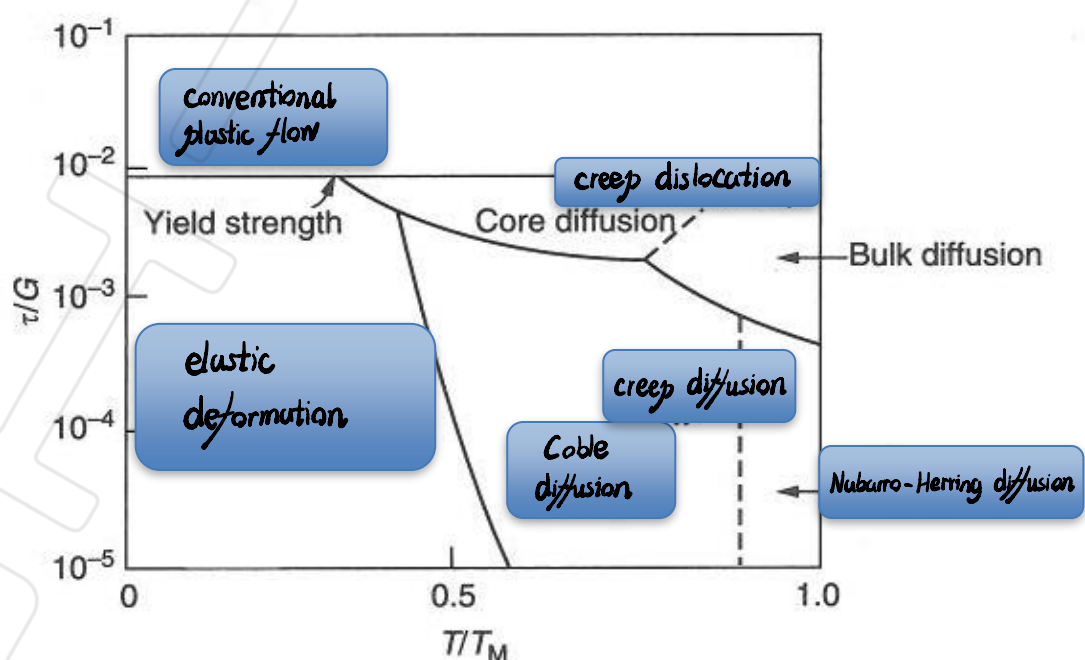
In order to maintain continuity within the creeping material, the grains must rearrange themselves and they can do so only by sliding along each other, which is grain boundary sliding.

Diffusion creep is influenced by the diffusion coefficient therefore stress, temperature and grain size.

$$\dot{\epsilon} = \frac{C \sigma e^{-\frac{q}{kT}}}{d^2}$$

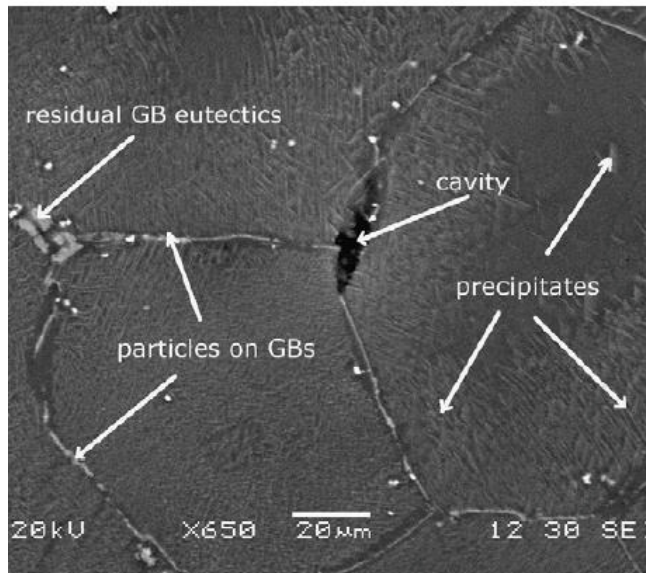
(d^2 for Nabarro-Herring creep)

4. Ashby maps



5. Application: Magnesium transmission housings design against creep.

Below are magnesium WE43 alloy SEM images. Comparing with the design against creep criteria table, explain what has been done in the WE43 alloy against creep?

**Design against dislocation creep**

- 1a** Choose a material with a high melting point (minimise T/T_m)
- 1b** Maximise obstacles to dislocation motion that are stable at operation T
- 1c** Choose an alloy where the majority phase has a high lattice resistance

Design against diffusion creep

- 2a** Choose a material with a high melting point (minimise T/T_m)
- 2b** Maximise the grain size to give long diffusion distances and little gb diffusion.
- 2c** Choose an alloy with precipitates at gb's that are stable at operation T to impede gb sliding